EXPERIMENT -3

AIM- Create an Employee–Department–Project schema. Insert at least 30 employees across 5 departments and 8 projects. Write SQL queries demonstrating selection, projection, aggregates, GROUP BY, HAVING, CASE expressions, and ORDER BY.

OBJECTIVES

- To create relational tables for Department, Employee, and Project.
- To establish relationships between the tables using Primary Keys and Foreign Keys.
- To insert at least 30 employees, 5 departments, and 8 projects.
- To retrieve specific records using Selection.
- To retrieve specific attributes using Projection.
- To perform calculations using aggregate functions such as COUNT(), AVG(), MAX(), MIN(), and SUM().
- To group records using GROUP BY.
- To filter grouped results using HAVING.
- To categorize employee data using a CASE expression.
- To sort query results using ORDER BY.

THEORY

DATABASE SCHEMA

A database schema defines the logical structure of a database, including its tables, attributes, and relationships. In this experiment, the database consists of the Department, Employee, Project, and Employee_Project relations. The Department relation stores department information, the Employee relation stores employee details, the Project relation stores project information, and the Employee_Project relation associates employees with projects. The database contains 5 departments, 8 projects, and 30 employees.

SELECTION

Selection is used to retrieve records from a relation that satisfy a specified condition. In SQL, selection is generally performed using the WHERE clause. It filters the rows of a table according to the given condition.

PROJECTION

Projection is used to retrieve only the required attributes or columns from a relation. In SQL, projection is performed using the SELECT statement by specifying the required columns. It allows only the relevant attributes to be displayed in the result.

AGGREGATE FUNCTIONS

Aggregate functions perform calculations on a set of rows and return a single result. The aggregate functions used in this experiment are:

- COUNT() – Counts the number of rows.
- SUM() – Calculates the total of numeric values.
- AVG() – Calculates the average of numeric values.
- MAX() – Returns the maximum value.
- MIN() – Returns the minimum value.

GROUP BY

The GROUP BY clause is used to arrange rows into groups based on one or more columns. It is commonly used with aggregate functions to perform calculations separately for each group. In this experiment, it is used for department-wise analysis of employees and salaries.

HAVING

The HAVING clause is used to apply conditions to groups created using the GROUP BY clause. It is generally used with aggregate functions to filter grouped results based on a specified condition.

CASE EXPRESSION

The CASE expression is used to perform conditional operations in SQL. It evaluates specified conditions and returns a corresponding value when a condition is satisfied. In this experiment, it is used to categorize employees based on their salary.

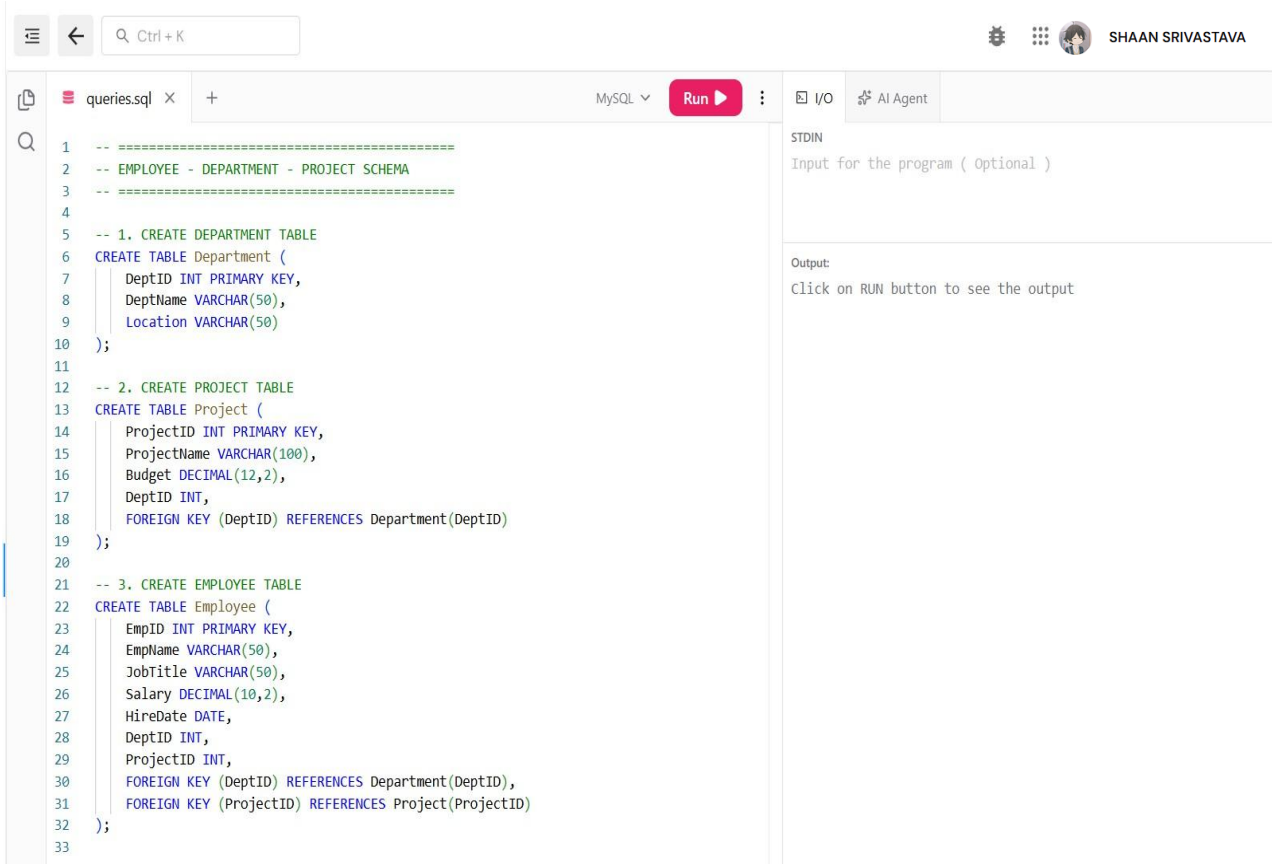
ORDER BY

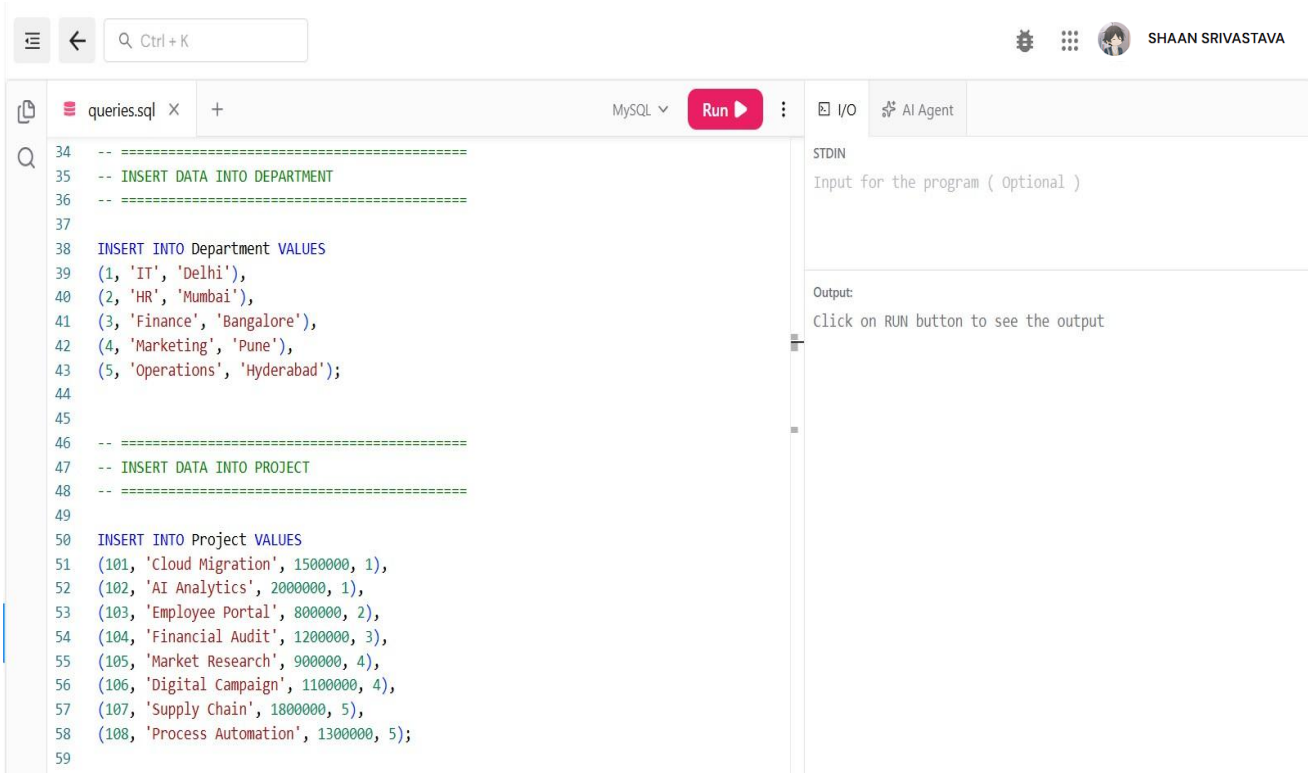
The ORDER BY clause is used to sort the records returned by a query. The results can be arranged in ascending order using ASC or descending order using DESC. In this experiment, it is used to arrange employees according to their salary.

DATABASE RELATIONSHIPS

The database represents the relationship between employees, departments, and projects. A department can have multiple employees and projects. An employee can work on multiple projects, and a project can have multiple employees. The Employee_Project relation is used to represent the association between employees and projects.

SOURCE CODE-





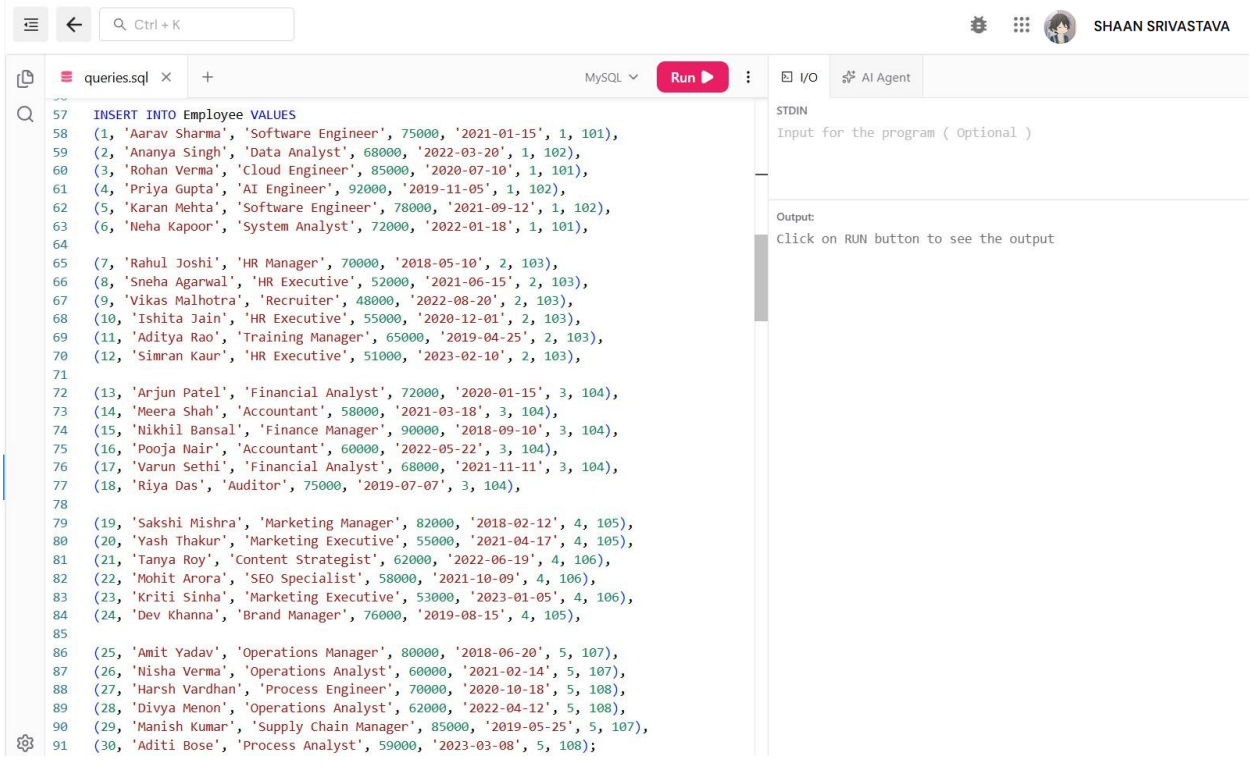
```sql
-- ============================================
-- INSERT DATA INTO DEPARTMENT
-- ============================================

INSERT INTO Department VALUES
(1, 'IT', 'Delhi'),
(2, 'HR', 'Mumbai'),
(3, 'Finance', 'Bangalore'),
(4, 'Marketing', 'Pune'),
(5, 'Operations', 'Hyderabad');


-- ============================================
-- INSERT DATA INTO PROJECT
-- ============================================

INSERT INTO Project VALUES
(101, 'Cloud Migration', 1500000, 1),
(102, 'AI Analytics', 2000000, 1),
(103, 'Employee Portal', 800000, 2),
(104, 'Financial Audit', 1200000, 3),
(105, 'Market Research', 900000, 4),
(106, 'Digital Campaign', 1100000, 4),
(107, 'Supply Chain', 1800000, 5),
(108, 'Process Automation', 1300000, 5);

```



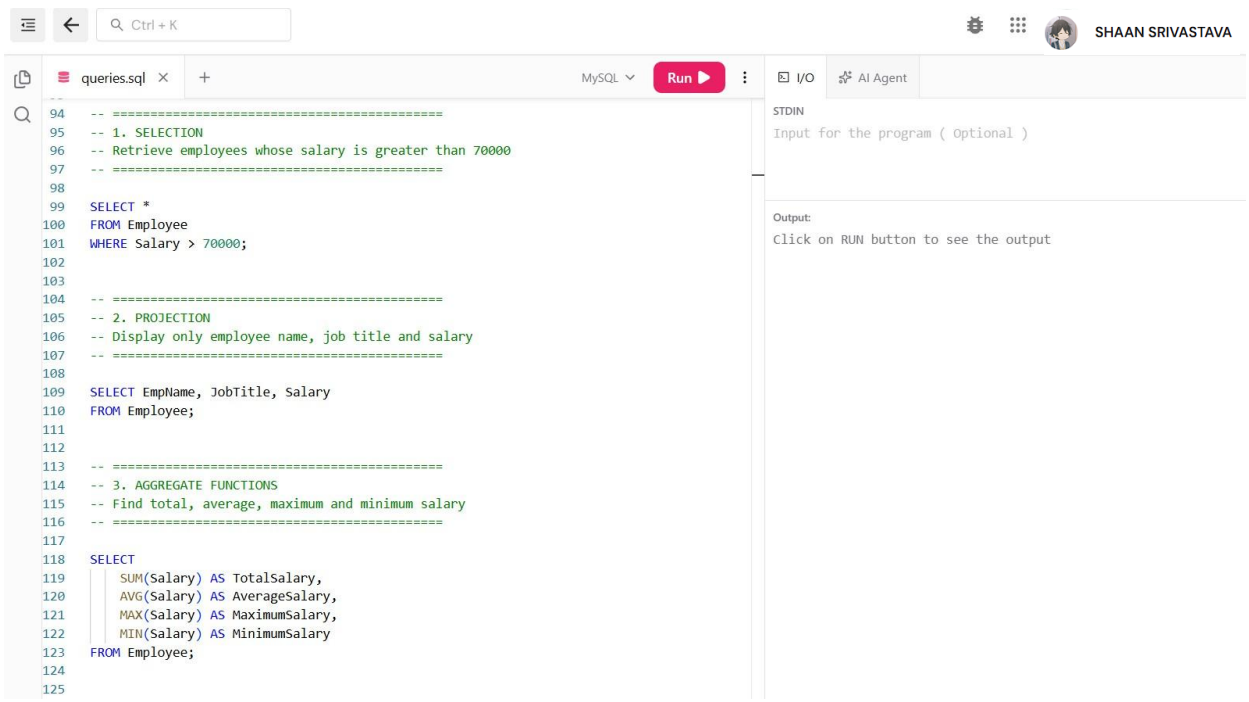
```sql
INSERT INTO Employee VALUES
(1, 'Aarav Sharma', 'Software Engineer', 75000, '2021-01-15', 1, 101),
(2, 'Ananya Singh', 'Data Analyst', 68000, '2022-03-20', 1, 102),
(3, 'Rohan Verma', 'Cloud Engineer', 85000, '2020-07-10', 1, 101),
(4, 'Priya Gupta', 'AI Engineer', 92000, '2019-11-05', 1, 102),
(5, 'Karan Mehta', 'Software Engineer', 78000, '2021-09-12', 1, 102),
(6, 'Neha Kapoor', 'System Analyst', 72000, '2022-01-18', 1, 101),

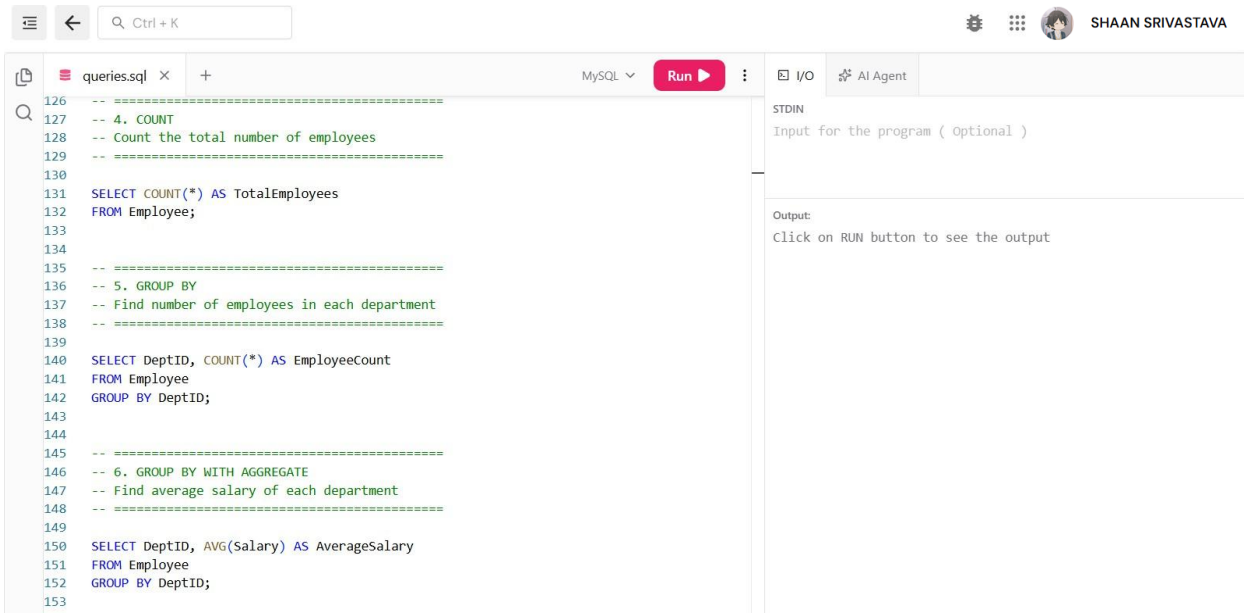
(7, 'Rahul Joshi', 'HR Manager', 70000, '2018-05-10', 2, 103),
(8, 'Sneha Agarwal', 'HR Executive', 52000, '2021-06-15', 2, 103),
(9, 'Vikas Malhotra', 'Recruiter', 48000, '2022-08-20', 2, 103),
(10, 'Ishita Jain', 'HR Executive', 55000, '2020-12-01', 2, 103),
(11, 'Aditya Rao', 'Training Manager', 65000, '2019-04-25', 2, 103),
(12, 'Simran Kaur', 'HR Executive', 51000, '2023-02-10', 2, 103),

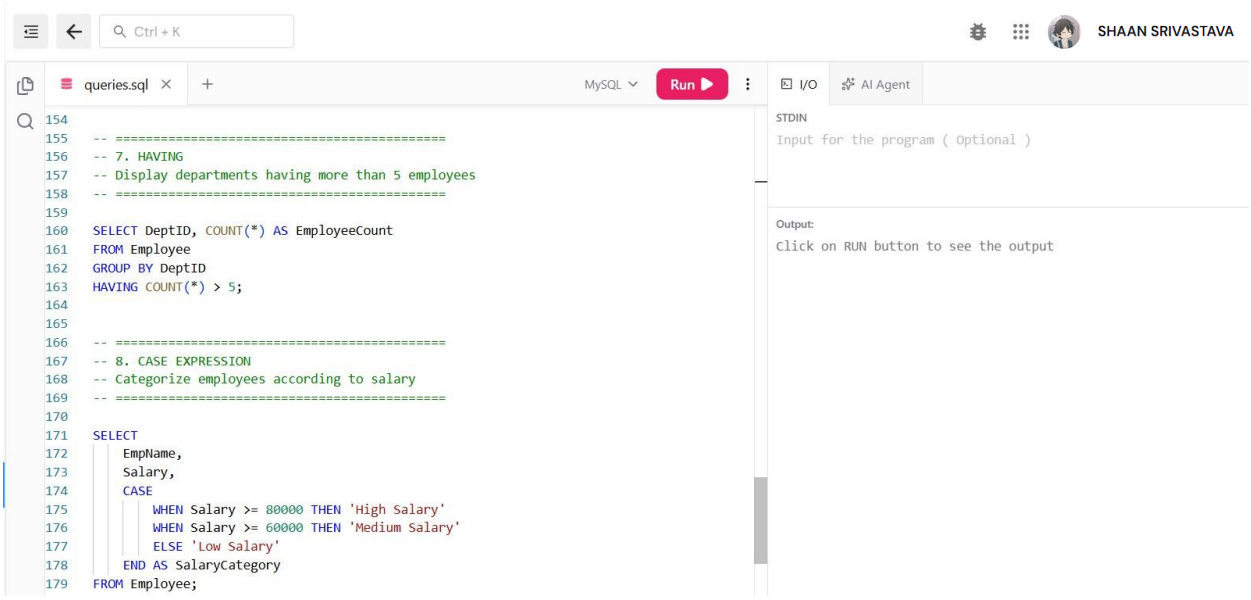
(13, 'Arjun Patel', 'Financial Analyst', 72000, '2020-01-15', 3, 104),
(14, 'Meera Shah', 'Accountant', 58000, '2021-03-18', 3, 104),
(15, 'Nikhil Bansal', 'Finance Manager', 90000, '2018-09-10', 3, 104),
(16, 'Pooja Nair', 'Accountant', 60000, '2022-05-22', 3, 104),
(17, 'Varun Sethi', 'Financial Analyst', 68000, '2021-11-11', 3, 104),
(18, 'Riya Das', 'Auditor', 75000, '2019-07-07', 3, 104),

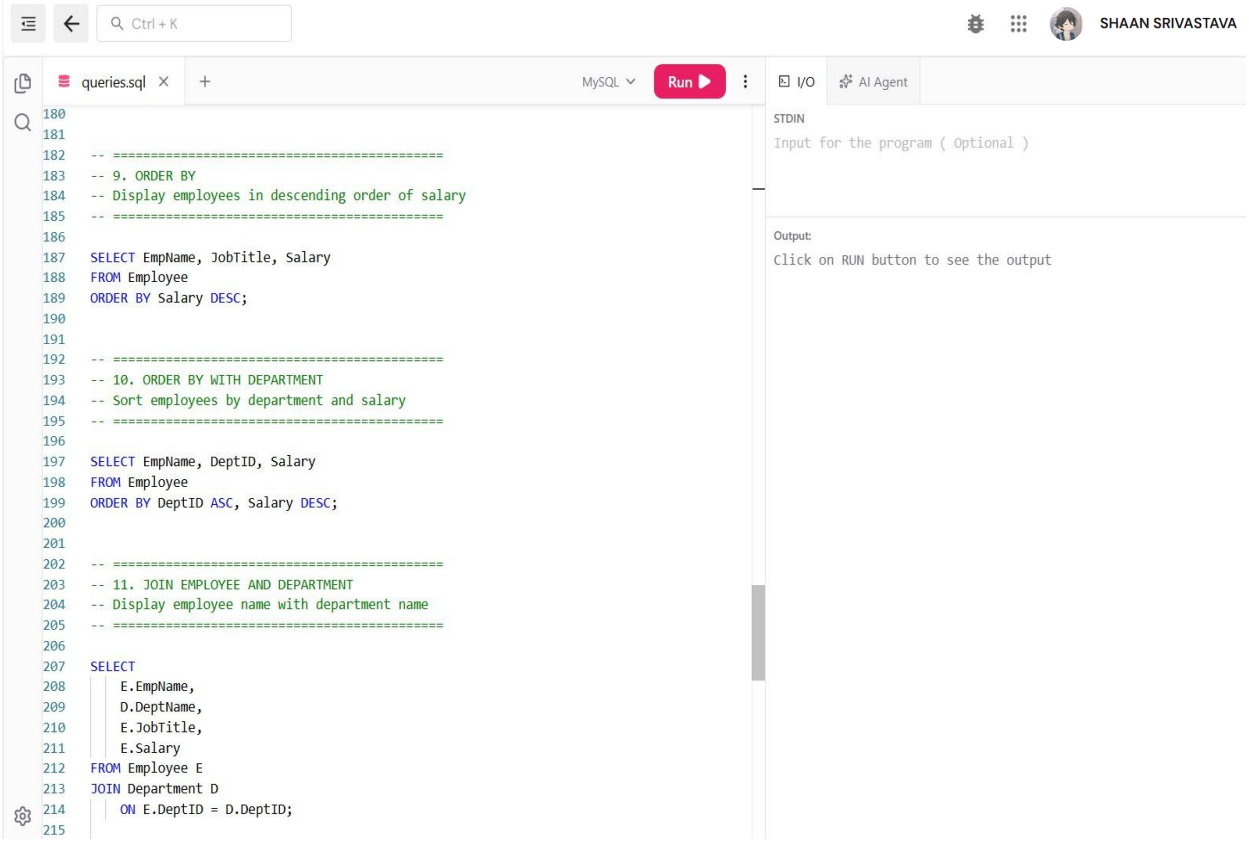
(19, 'Sakshi Mishra', 'Marketing Manager', 82000, '2018-02-12', 4, 105),
(20, 'Yash Thakur', 'Marketing Executive', 55000, '2021-04-17', 4, 105),
(21, 'Tanya Roy', 'Content Strategist', 62000, '2022-06-19', 4, 106),
(22, 'Mohit Arora', 'SEO Specialist', 58000, '2021-10-09', 4, 106),
(23, 'Kriti Sinha', 'Marketing Executive', 53000, '2023-01-05', 4, 106),
(24, 'Dev Khanna', 'Brand Manager', 76000, '2019-08-15', 4, 105),

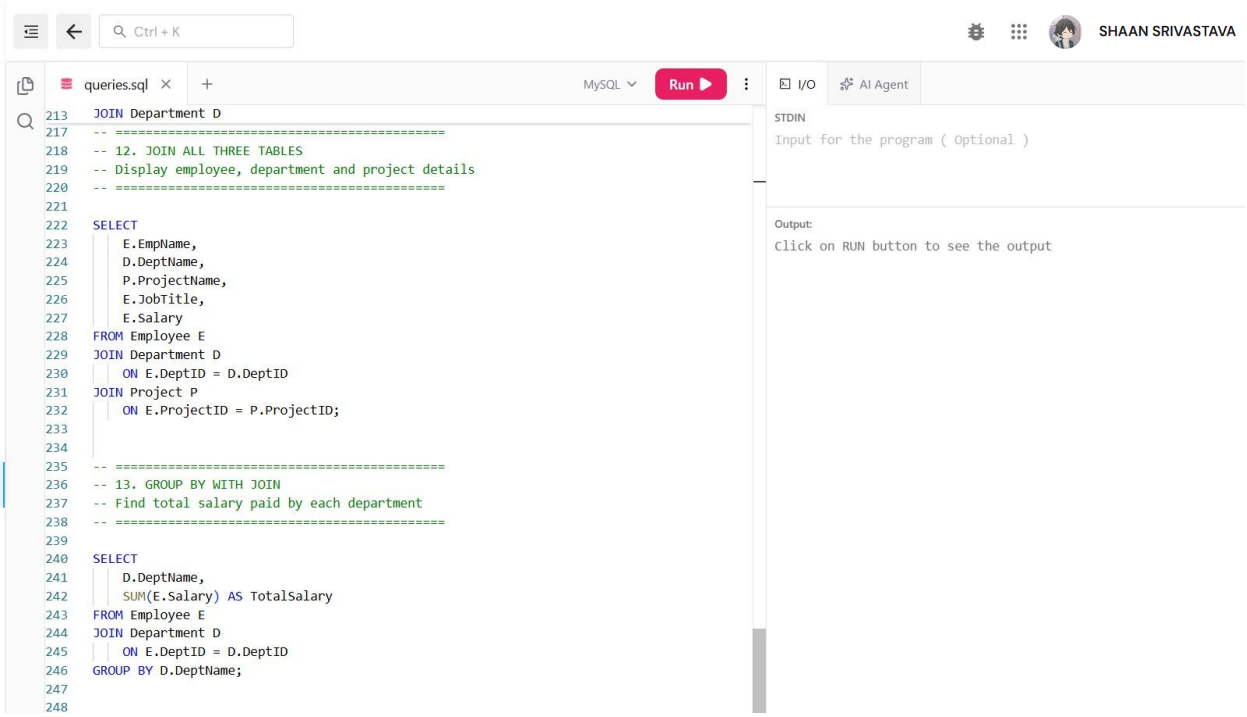
(25, 'Amit Yadav', 'Operations Manager', 80000, '2018-06-20', 5, 107),
(26, 'Nisha Verma', 'Operations Analyst', 60000, '2021-02-14', 5, 107),
(27, 'Harsh Vardhan', 'Process Engineer', 70000, '2020-10-18', 5, 108),
(28, 'Divya Menon', 'Operations Analyst', 62000, '2022-04-12', 5, 108),
(29, 'Manish Kumar', 'Supply Chain Manager', 85000, '2019-05-25', 5, 107),
(30, 'Aditi Bose', 'Process Analyst', 59000, '2023-03-08', 5, 108);
```


Ctrl + K
SHAAN SRIVASTAVA
queries.sql
MySQL
Run
I/O
AI Agent
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
-- ==
-- 1. SELECTION
-- Retrieve employees whose salary is greater than 70000
-- ==
SELECT *
FROM Employee
WHERE Salary > 70000;
-- ==
-- 2. PROJECTION
-- Display only employee name, job title and salary
-- ==
SELECT EmpName, JobTitle, Salary
FROM Employee;
-- ==
-- 3. AGGREGATE FUNCTIONS
-- Find total, average, maximum and minimum salary
-- ==
SELECT
SUM(Salary) AS TotalSalary,
AVG(Salary) AS AverageSalary,
MAX(Salary) AS MaximumSalary,
MIN(Salary) AS MinimumSalary
FROM Employee;

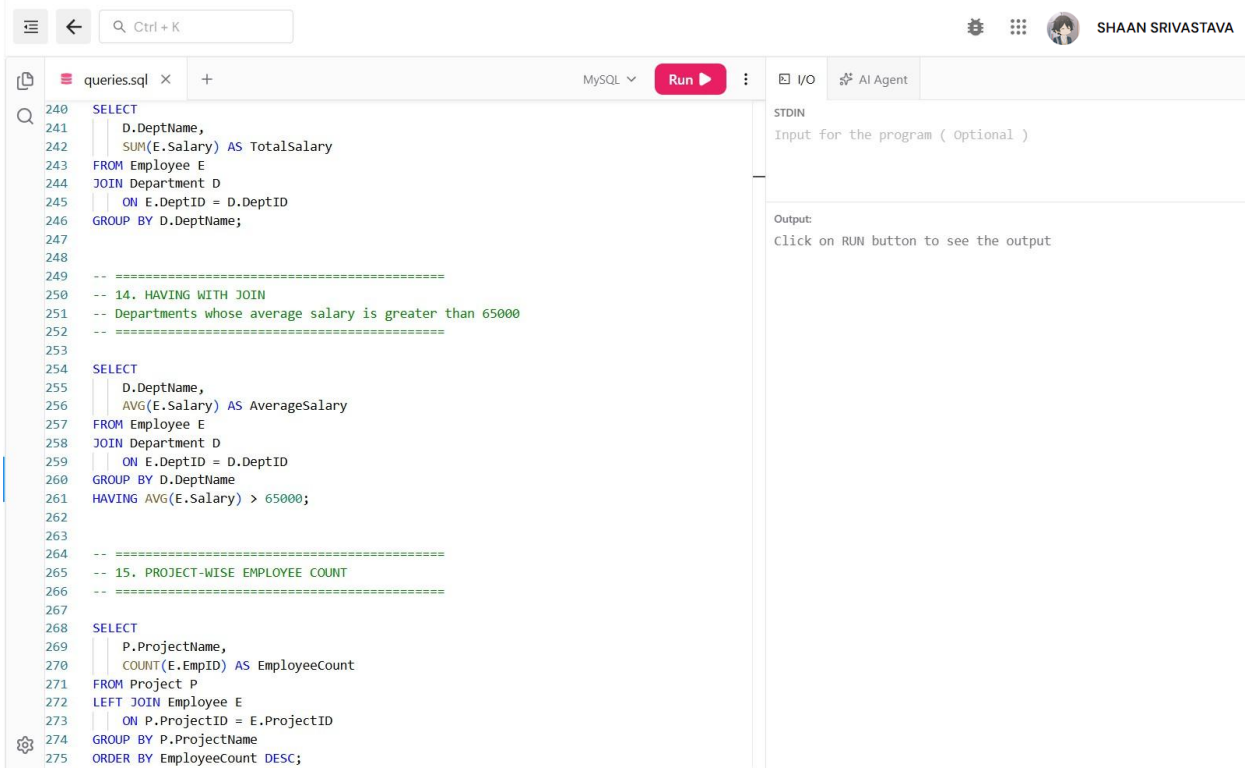

Ctrl + K
SHAAN SRIVASTAVA
queries.sql
MySQL
Run
I/O
AI Agent
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
-- ==
-- 4. COUNT
-- Count the total number of employees
-- ==
SELECT COUNT(*) AS TotalEmployees
FROM Employee;
-- ==
-- 5. GROUP BY
-- Find number of employees in each department
-- ==
SELECT DeptID, COUNT(*) AS EmployeeCount
FROM Employee
GROUP BY DeptID;
-- ==
-- 6. GROUP BY WITH AGGREGATE
-- Find average salary of each department
-- ==
SELECT DeptID, AVG(Salary) AS AverageSalary
FROM Employee
GROUP BY DeptID;


Ctrl + K
SHAAN SRIVASTAVA
queries.sql
MySQL
Run
I/O
AI Agent
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
-- ==========
-- 7. HAVING
-- Display departments having more than 5 employees
-- ==========
SELECT DeptID, COUNT(*) AS EmployeeCount
FROM Employee
GROUP BY DeptID
HAVING COUNT(*) > 5;
-- ==========
-- 8. CASE EXPRESSION
-- Categorize employees according to salary
-- ==========
SELECT
EmpName,
Salary,
CASE
WHEN Salary >= 80000 THEN 'High Salary'
WHEN Salary >= 60000 THEN 'Medium Salary'
ELSE 'Low Salary'
END AS SalaryCategory
FROM Employee;


Ctrl + K
SHAAN SRIVASTAVA
queries.sql
MySQL
Run
I/O
AI Agent
STDIN
Input for the program (Optional)
Output:
Click on RUN button to see the output
-- ==========
-- 9. ORDER BY
-- Display employees in descending order of salary
-- ==========
SELECT EmpName, JobTitle, Salary
FROM Employee
ORDER BY Salary DESC;
-- ==========
-- 10. ORDER BY WITH DEPARTMENT
-- Sort employees by department and salary
-- ==========
SELECT EmpName, DeptID, Salary
FROM Employee
ORDER BY DeptID ASC, Salary DESC;
-- ==========
-- 11. JOIN EMPLOYEE AND DEPARTMENT
-- Display employee name with department name
-- ==========
SELECT
E.EmpName,
D.DeptName,
E.JobTitle,
E.Salary
FROM Employee E
JOIN Department D
ON E.DeptID = D.DeptID;




```
213  JOIN Department D
217  -- ============================================
218  -- 12. JOIN ALL THREE TABLES
219  -- Display employee, department and project details
220  -- ============================================
221
222  SELECT
223      E.EmpName,
224      D.DeptName,
225      P.ProjectName,
226      E.JobTitle,
227      E.Salary
228  FROM Employee E
229  JOIN Department D
230      ON E.DeptID = D.DeptID
231  JOIN Project P
232      ON E.ProjectID = P.ProjectID;
233
234
235  -- ============================================
236  -- 13. GROUP BY WITH JOIN
237  -- Find total salary paid by each department
238  -- ============================================
239
240  SELECT
241      D.DeptName,
242      SUM(E.Salary) AS TotalSalary
243  FROM Employee E
244  JOIN Department D
245      ON E.DeptID = D.DeptID
246  GROUP BY D.DeptName;
247
248
```




```
240  SELECT
241      D.DeptName,
242      SUM(E.Salary) AS TotalSalary
243  FROM Employee E
244  JOIN Department D
245      ON E.DeptID = D.DeptID
246  GROUP BY D.DeptName;
247
248
249  -- ============================================
250  -- 14. HAVING WITH JOIN
251  -- Departments whose average salary is greater than 65000
252  -- ============================================
253
254  SELECT
255      D.DeptName,
256      AVG(E.Salary) AS AverageSalary
257  FROM Employee E
258  JOIN Department D
259      ON E.DeptID = D.DeptID
260  GROUP BY D.DeptName
261  HAVING AVG(E.Salary) > 65000;
262
263
264  -- ============================================
265  -- 15. PROJECT-WISE EMPLOYEE COUNT
266  -- ============================================
267
268  SELECT
269      P.ProjectName,
270      COUNT(E.EmpID) AS EmployeeCount
271  FROM Project P
272  LEFT JOIN Employee E
273      ON P.ProjectID = E.ProjectID
274  GROUP BY P.ProjectName
275  ORDER BY EmployeeCount DESC;
```

OUTPUT

SHAAN SRIVASTAVA

I/O | AI Agent | 104 ms

STDIN

Input for the program (Optional)

Output:

```
+-------+---------------+----------------------+----------+------------+--------+-----------+
| EmpID | EmpName       | JobTitle             | Salary   | HireDate   | DeptID | ProjectID |
+-------+---------------+----------------------+----------+------------+--------+-----------+
|     1 | Aarav Sharma  | Software Engineer    | 75000.00 | 2021-01-15 |      1 |       101 |
|     3 | Rohan Verma   | Cloud Engineer       | 85000.00 | 2020-07-10 |      1 |       101 |
|     4 | Priya Gupta   | AI Engineer          | 92000.00 | 2019-11-05 |      1 |       102 |
|     5 | Karan Mehta   | Software Engineer    | 78000.00 | 2021-09-12 |      1 |       102 |
|     6 | Neha Kapoor   | System Analyst       | 72000.00 | 2022-01-18 |      1 |       101 |
|    13 | Arjun Patel   | Financial Analyst    | 72000.00 | 2020-01-15 |      3 |       104 |
|    15 | Nikhil Bansal | Finance Manager      | 90000.00 | 2018-09-10 |      3 |       104 |
|    18 | Riya Das      | Auditor              | 75000.00 | 2019-07-07 |      3 |       104 |
|    19 | Sakshi Mishra | Marketing Manager    | 82000.00 | 2018-02-12 |      4 |       105 |
|    24 | Dev Khanna    | Brand Manager        | 76000.00 | 2019-08-15 |      4 |       105 |
|    25 | Amit Yadav    | Operations Manager   | 80000.00 | 2018-06-20 |      5 |       107 |
|    29 | Manish Kumar  | Supply Chain Manager | 85000.00 | 2019-05-25 |      5 |       107 |
+-------+---------------+----------------------+----------+------------+--------+-----------+
+---------------+----------------------+----------+
| EmpName       | JobTitle             | Salary   |
+---------------+----------------------+----------+
| Aarav Sharma  | Software Engineer    | 75000.00 |
| Ananya Singh  | Data Analyst         | 68000.00 |
| Rohan Verma   | Cloud Engineer       | 85000.00 |
| Priya Gupta   | AI Engineer          | 92000.00 |
| Karan Mehta   | Software Engineer    | 78000.00 |
| Neha Kapoor   | System Analyst       | 72000.00 |
| Rahul Joshi   | HR Manager           | 70000.00 |
| Sneha Agarwal | HR Executive         | 52000.00 |
| Vikas Malhotra| Recruiter            | 48000.00 |
```

SHAAN SRIVASTAVA

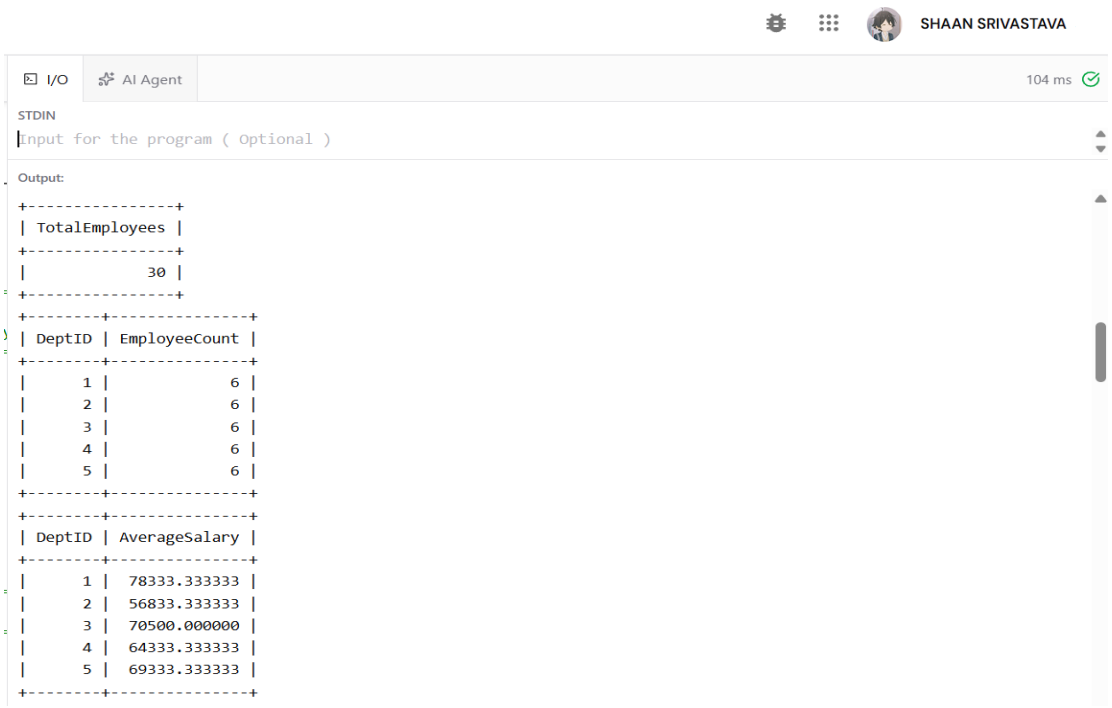
I/O | AI Agent | 104 ms

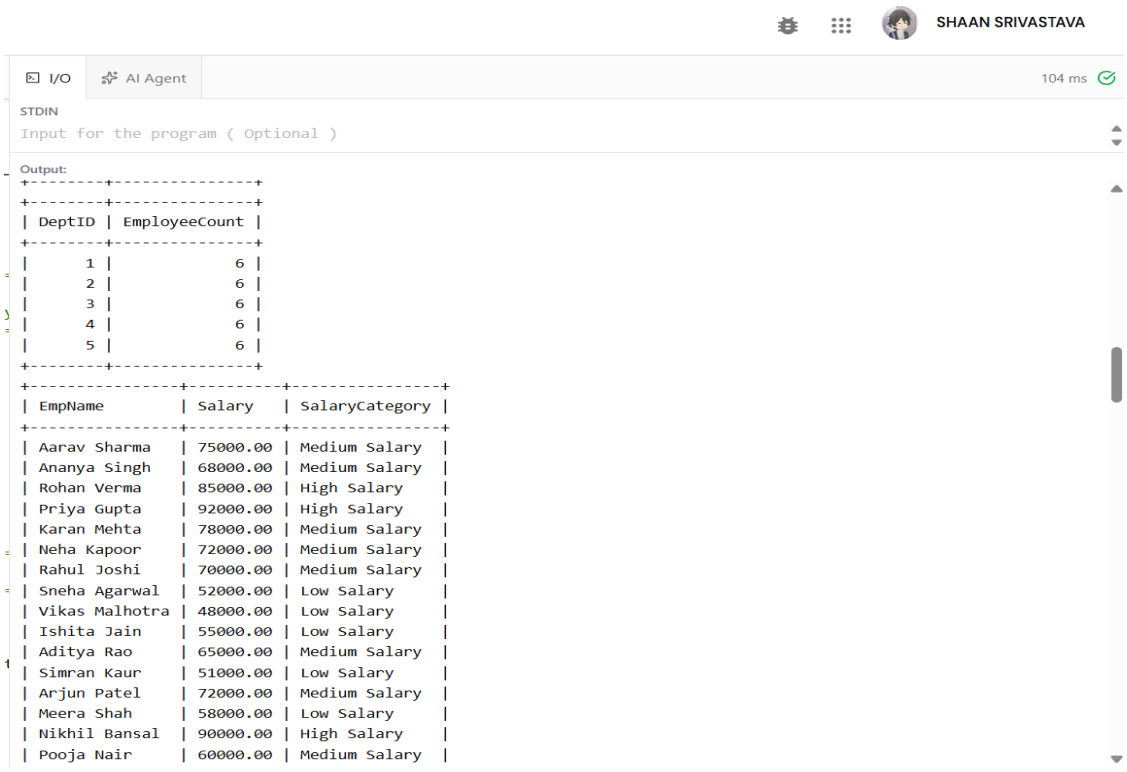
STDIN

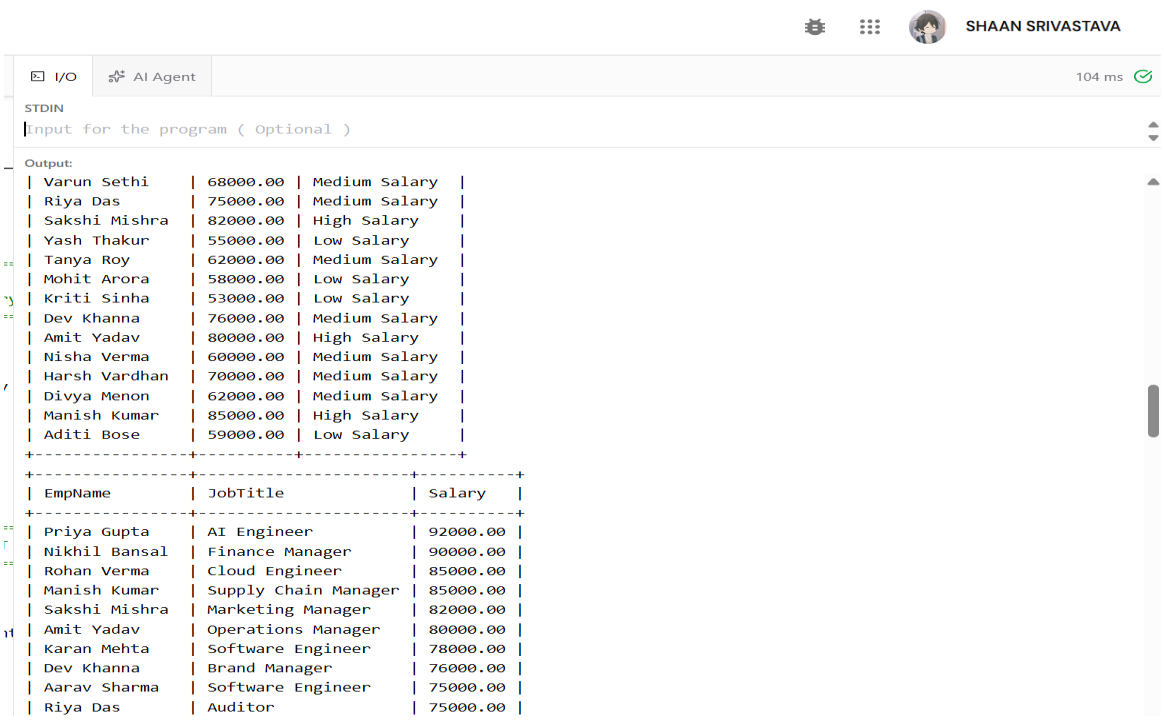
Input for the program (Optional)

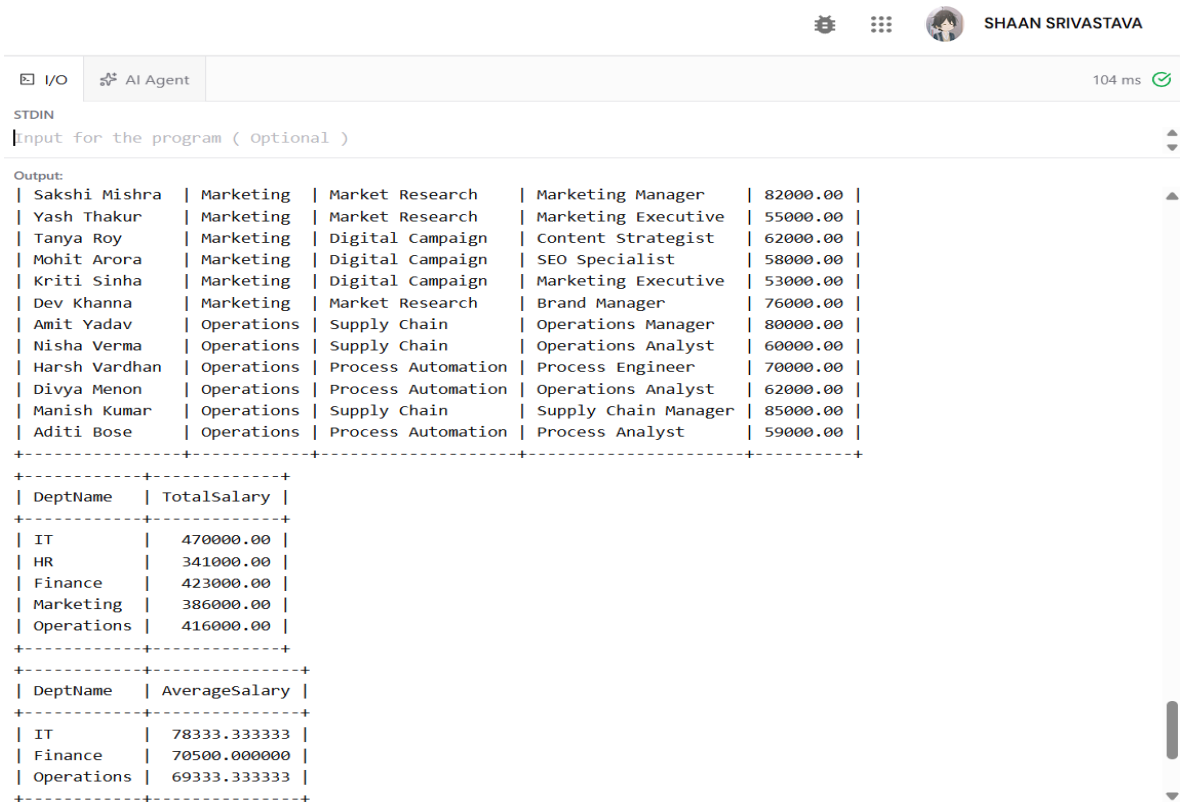
Output:

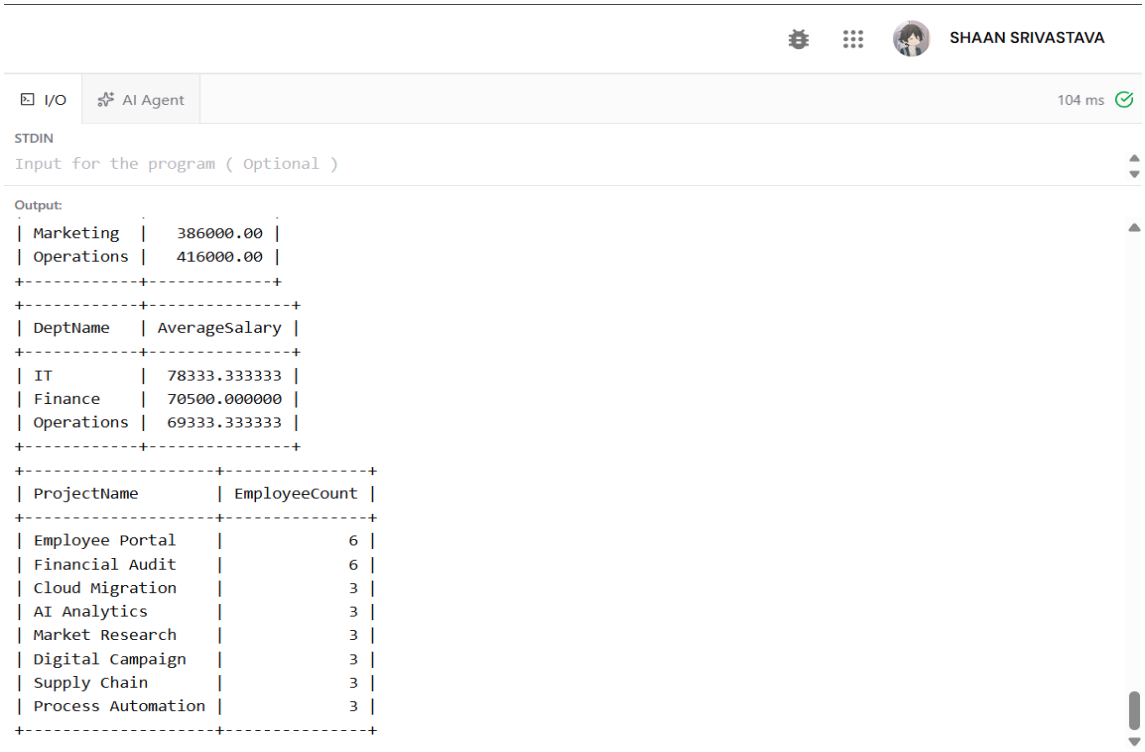
```
| Ishita Jain    | HR Executive         | 55000.00 |
| Aditya Rao     | Training Manager     | 65000.00 |
| Simran Kaur    | HR Executive         | 51000.00 |
| Arjun Patel    | Financial Analyst    | 72000.00 |
| Meera Shah     | Accountant           | 58000.00 |
| Nikhil Bansal  | Finance Manager      | 90000.00 |
| Pooja Nair     | Accountant           | 60000.00 |
| Varun Sethi    | Financial Analyst    | 68000.00 |
| Riya Das       | Auditor              | 75000.00 |
| Sakshi Mishra  | Marketing Manager    | 82000.00 |
| Yash Thakur    | Marketing Executive  | 55000.00 |
| Tanya Roy      | Content Strategist   | 62000.00 |
| Mohit Arora    | SEO Specialist       | 58000.00 |
| Kriti Sinha    | Marketing Executive  | 53000.00 |
| Dev Khanna     | Brand Manager        | 76000.00 |
| Amit Yadav     | Operations Manager   | 80000.00 |
| Nisha Verma    | Operations Analyst   | 60000.00 |
| Harsh Vardhan  | Process Engineer     | 70000.00 |
| Divya Menon    | Operations Analyst   | 62000.00 |
| Manish Kumar   | Supply Chain Manager | 85000.00 |
| Aditi Bose     | Process Analyst      | 59000.00 |
+----------------+----------------------+----------+
+-------------+---------------+---------------+---------------+
| TotalSalary | AverageSalary | MaximumSalary | MinimumSalary |
+-------------+---------------+---------------+---------------+
|  2036000.00 |  67866.666667 |      92000.00 |      48000.00 |
+-------------+---------------+---------------+---------------+
```


SHAAN SRIVASTAVA
I/O
AI Agent
104 ms
STDIN
Input for the program (Optional)
Output:
+----------------+
| TotalEmployees |
+----------------+
| 30 |
+----------------+
+---------+---------------+
| DeptID | EmployeeCount |
+---------+---------------+
1	6
2	6
3	6
4	6
5	6
+---------+---------------+
+---------+---------------+
| DeptID | AverageSalary |
+---------+---------------+
1	78333.333333
2	56833.333333
3	70500.000000
4	64333.333333
5	69333.333333
+---------+---------------+


SHAAN SRIVASTAVA
I/O
AI Agent
104 ms
STDIN
Input for the program (Optional)
Output:
+---------+---------------+
+---------+---------------+
| DeptID | EmployeeCount |
+---------+---------------+
1	6
2	6
3	6
4	6
5	6
+---------+---------------+
+-----------------+----------+----------------+
| EmpName | Salary | SalaryCategory |
+-----------------+----------+----------------+
Aarav Sharma	75000.00	Medium Salary
Ananya Singh	68000.00	Medium Salary
Rohan Verma	85000.00	High Salary
Priya Gupta	92000.00	High Salary
Karan Mehta	78000.00	Medium Salary
Neha Kapoor	72000.00	Medium Salary
Rahul Joshi	70000.00	Medium Salary
Sneha Agarwal	52000.00	Low Salary
Vikas Malhotra	48000.00	Low Salary
Ishita Jain	55000.00	Low Salary
Aditya Rao	65000.00	Medium Salary
Simran Kaur	51000.00	Low Salary
Arjun Patel	72000.00	Medium Salary
Meera Shah	58000.00	Low Salary
Nikhil Bansal	90000.00	High Salary
Pooja Nair	60000.00	Medium Salary


SHAAN SRIVASTAVA
I/O
AI Agent
104 ms
STDIN
Input for the program (Optional)
Output:
Varun Sethi	68000.00	Medium Salary
Riya Das	75000.00	Medium Salary
Sakshi Mishra	82000.00	High Salary
Yash Thakur	55000.00	Low Salary
Tanya Roy	62000.00	Medium Salary
Mohit Arora	58000.00	Low Salary
Kriti Sinha	53000.00	Low Salary
Dev Khanna	76000.00	Medium Salary
Amit Yadav	80000.00	High Salary
Nisha Verma	60000.00	Medium Salary
Harsh Vardhan	70000.00	Medium Salary
Divya Menon	62000.00	Medium Salary
Manish Kumar	85000.00	High Salary
Aditi Bose	59000.00	Low Salary
EmpName	JobTitle	Salary
Priya Gupta	AI Engineer	92000.00
Nikhil Bansal	Finance Manager	90000.00
Rohan Verma	Cloud Engineer	85000.00
Manish Kumar	Supply Chain Manager	85000.00
Sakshi Mishra	Marketing Manager	82000.00
Amit Yadav	Operations Manager	80000.00
Karan Mehta	Software Engineer	78000.00
Dev Khanna	Brand Manager	76000.00
Aarav Sharma	Software Engineer	75000.00
Riya Das	Auditor	75000.00


SHAAN SRIVASTAVA
I/O
AI Agent
104 ms
STDIN
Input for the program (Optional)
Output:
Sakshi Mishra	Marketing	Market Research	Marketing Manager	82000.00
Yash Thakur	Marketing	Market Research	Marketing Executive	55000.00
Tanya Roy	Marketing	Digital Campaign	Content Strategist	62000.00
Mohit Arora	Marketing	Digital Campaign	SEO Specialist	58000.00
Kriti Sinha	Marketing	Digital Campaign	Marketing Executive	53000.00
Dev Khanna	Marketing	Market Research	Brand Manager	76000.00
Amit Yadav	Operations	Supply Chain	Operations Manager	80000.00
Nisha Verma	Operations	Supply Chain	Operations Analyst	60000.00
Harsh Vardhan	Operations	Process Automation	Process Engineer	70000.00
Divya Menon	Operations	Process Automation	Operations Analyst	62000.00
Manish Kumar	Operations	Supply Chain	Supply Chain Manager	85000.00
Aditi Bose	Operations	Process Automation	Process Analyst	59000.00
DeptName	TotalSalary
IT	470000.00
HR	341000.00
Finance	423000.00
Marketing	386000.00
Operations	416000.00
DeptName	AverageSalary
IT	78333.333333
Finance	70500.000000
Operations	69333.333333



OBSERVATION/OUTCOME

1. Employee, Department, and Project tables were successfully created.
2. Data for 30 employees, 5 departments, and 8 projects was successfully inserted.
3. Selection and projection queries successfully retrieved the required records and attributes.
4. Aggregate functions such as COUNT(), SUM(), AVG(), MAX(), and MIN() were successfully executed.
5. GROUP BY and HAVING were used to group and filter grouped data.
6. CASE expression successfully categorized employees based on salary.
7. ORDER BY successfully sorted employee records according to the specified conditions.
8. Queries using joins successfully retrieved related data from multiple tables.